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Effects of prescribed fire on habitat of beaver (*Castor canadensis*) in Elk Island National Park, Canada

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Abstract

Fire, flooding, herbivory, and the effects of climate are all topical issues for today's land managers. Effective resource management requires a balance among these processes, which in turn, requires a better understanding of their interactions. Beaver (*Castor canadensis*) are strong colonizers and have been successfully reintroduced to much of their former range. Prescribed fire has also been introduced in many areas as a management tool to restore ecological function. Resource managers have often assumed fire would also benefit non-target species like beaver; however, its effect on beaver has not been well studied. In this study, part of a broader project in Elk Island National Park, Canada, we examine the effect of prescribed fire on beaver lodge occupancy in the context of high ungulate populations. Elk Island National Park has an active beaver population, high ungulate densities, and a well-established prescribed fire program. We examine whether frequency, size, and timing of burns influence beaver lodge occupancy and the establishment of new lodges. Since 1979, over 51% of the park (99.3 km^2) has been burned with prescribed fire. By comparing lodge occupancy over a period prior to and after a series of prescribed burns, we analyzed beaver occupancy rates pre- and post-burn. Our results show that repeated burning dramatically decreases beaver lodge occupancy, and that even after one burn the number of active colonies declines and does not recover to pre-fire populations. Especially when combined with drought and herbivory, prescribed fire does not improve beaver habitat.

Keywords: Beaver; Beaver habitat; *Castor canadensis*; Prescribed fire; Elk Island National Park; Fire management; Ungulates; Herbivory; Aspen Parkland; Boreal mixed-wood forest; Drought; Climate

1. Introduction

Flooding by beaver (*Castor canadensis*) and the effects of fire are viewed as major natural landscape processes that drive vegetation dynamics and alter ecosystems at both local and regional scales (Naiman et al., 1988; Johnston and Naiman, 1990a; Ford et al., 1999; Bisson et al., 2003). So important is the beaver's role in maintaining wetlands on the landscape that in many areas in North America beaver have been reintroduced to former ranges (Naiman et al., 1988; Wright et al., 2002). The beaver's ability to impound large areas of water, alter vegetation communities, and change sediment loads and water chemistry is well documented (Naiman et al., 1986; Smith et al., 1991; Klotz, 1998). For separate management objectives, fire has also been reintroduced in many places where fire suppression, landscape fragmentation, and climate have altered historic fire regimes and vegetation communities (Bartos and Mueggler, 1979; White et al., 1998; Ford et al., 1999; Weir et al., 2000). The interaction between fire and beaver has been assumed to be beneficial to beaver (Kellyhouse, 1979; Naiman et al., 1988), although this assumption has rarely been tested.

Prescribed fire, also referred to as ‘‘ecological burning’’ (Meers and Adams, 2003), is often used to restore grasslands (Bailey et al., 1990; Roques et al., 2001) and stimulate growth of woody plants (Bartos and Mueggler, 1979; Elliot et al., 1999; Hiers et al., 2003). Use of fire to manipulate vegetation communities can also benefit various wildlife species by improving forage quality (Vinton et al., 1993) and habitat structure (Mushinsky, 1985). A growing body of literature however, reveals that, in the context of other ecological disturbances such as high herbivory and climate (e.g., drought), use of prescribed fire does not always achieve expected

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outcomes (White et al., 1998; Hessl and Graumlich, 2002). These findings suggest that land managers should adapt their fire management practices to address ecosystem dynamics.

Although one of the preferred forage species for beaver, trembling aspen (*Populus tremuloides*), regenerates well after fire (Bartos and Mueggler, 1979; Bartos and Mueggler, 1981; Campbell et al., 1994; Bailey and Whitham, 2002), herbivory by ungulates can significantly suppress or reduce ramet production (Campbell et al., 1994; White et al., 1998; Bailey and Whitham, 2002; Hessl and Graumlich, 2002; White et al., 2003). Combined with fire, high levels of ungulate herbivory can dramatically reduce regeneration of woody species (Bailey et al., 1990; Bork et al., 1997a; White et al., 1998). Bailey and Whitham (2002) found that aspen stands exposed to high-severity burns but no elk (*Cervus elaphus*) had 10 times greater aboveground biomass than stands exposed to intermediate burns. The combination of elk and high-severity burns, however, reduced above-ground biomass 90-fold.

In the Rocky Mountain national parks of Canada and the United States the use of prescribed fire on landscapes containing high elk densities was found to hinder aspen regeneration (White et al., 1998, 2003). Browsing by elk impeded the rejuvenation of aspen stems following prescribed fire. Aspen stems were browsed before they could reach 2-m in height. Meers and Adams (2003) found a similar situation in southeastern Australia where high levels of preferential grazing by eastern kangaroo (*Macopus giganteus*) became so concentrated in areas burned with prescribed fire that there were significantly fewer shrubs in burned areas than unburned areas. They also predicted that because habitat condition is closely tied to precipitation, in drought years, grazing pressure after fire could result in even more severe impacts on the vegetation.

Little or no empirical evidence has been published on the beneficial effects of fire on beaver habitat, although anecdotal evidence has been presented in the literature (Bird, 1961; Kellyhouse, 1979; Lewis, 1982; Naiman et al., 1988; Fryxell, 2001). Logically, regeneration of woody species in riparian areas should provide additional forage for beaver. Higher soil moisture in riparian areas can provide some protection for plant roots and belowground biomass during fires. Many riparian shrubs and forbs recover well after fire (Dwire and Kauffman, 2003) and, for some species, reproductive output following a fire increases dramatically (Kaufmann and Martin, 1990; Pendergrass et al., 1999). Climate, it seems, is an overriding determinant of how vegetation and wildlife respond to fire (Weir et al., 2000).

However, as seen in the preceding studies, burning in areas with a high level of herbivory does not necessarily result in an increase in woody plants. In addition, during prescribed burns, most vegetation around beaver ponds is either burned or scorched. Morgan (1991) noted that beaver avoided trees on which the lower trunks had been singed or charred. An ethnographic study of traditional aboriginal use of fire in the Boreal forest reported that beavers did not return to a burned area until four years after the fire (Lewis, 1982).

Considering the importance of climate in prescribed fire in typical years, fire occurring during periods of drought could severely compromise the survival of woody plants and dramatically alter vegetation communities. Roques et al. (2001) found that combined with drought, fire significantly reduced shrub densities. Fire frequency also determines how plant communities will respond to fire. Elliot et al. (1999) determined that repeated fires would be necessary to regenerate declining pitch pine (*Pinus rigida*) stands in the southern Appalachians, while Roques et al. (2001) found that frequent fires in the context of low grazing pressures prevented shrub encroachment into the eastern region of Swaziland, southern Africa.

Fire severity, the depth of burn (de Groot et al., 2003), degree of removal of organic material and extent of soil heating during a fire (Schimmel and Granström, 1996), can have both immediate and lingering impacts on regeneration of forest vegetation. Growth of aspen suckers in the first year following a fire is negatively correlated with fire severity, but that effect does not persist in the second and third years following a burn (Wang, 2003). However, less fire tolerant species are not as resilient and are likely to be replaced by better-adapted species (de Groot et al., 2003). Therefore, severe or deep burning fires can cause changes in plant community composition and variation in re-growth of vegetation (Miyanishi, 2001). Simulations of future fire regimes in Canadian boreal forests predict fire severity would intensify with increased drought associated with a warming climate (de Groot et al., 2003).

In some areas of western Canada, such as Elk Island National Park, prescribed fires are usually conducted in the early spring or late fall, when higher soil moisture limits the spread of fire and offers better control of the fire. In dry conditions, fire can extend to the water's edge (personal observation), fed by dead vegetation from the past season. Where beaver occupy these wetlands, lodge and dams can be damaged or destroyed. With no food and limited escape cover, beaver may attempt to relocate to more suitable habitat. In such instances, beaver survival could be reduced further by predation risk associated with travel through open, burned areas.

Given the complex interactions between fire, climate, and herbivory, Elk Island National Park (EINP) in Alberta, Canada is an ideal place to examine the combined effects of these processes on beaver populations in a managed landscape. Park management conducted the first prescribed burn in the park in 1979. By 2002, over 51% of EINP had been burned at least once (in some areas as many as 8 times). In addition, the park has very high densities of large ungulates, a well-established beaver population, and has recently experienced severe drought. Together, prescribed fire, ungulate herbivory, and flooding by beaver have shaped EINP into a complex mosaic of wetlands, grasslands, southern boreal mixed-wood, and aspen forests (Campbell et al., 1994). Their influences on one another, separate or in combination, are less obvious. In particular, the effect of fire and ungulate herbivory on the park's beaver populations is not well understood. This paper specifically examines the effects of prescribed fire on beaver lodge occupancy in EINP, one component of a larger study addressing fire, herbivory and climate.

The purpose of this study is to examine whether prescribed fire influences beaver lodge occupancy differently under normal conditions compared to conditions of extreme drought, within the context of high levels of ungulate herbivory. This study evaluates changes in beaver lodge occupancy due to the predicted effects of fire on beaver habitat. Our objectives were to determine (1) whether beaver lodge occupancy is lower in areas that have been burned than in unburned areas over and above differences in wetland availability, (2) whether a decrease in beaver lodge occupancy in burned areas results in a predictable reoccupation of these areas over time, (3) whether fire frequency and time since the last fire (fire cycle) affects lodge abandonment, and (4) whether distances between abandoned and active lodges differ between burned and unburned habitats (such increases in distance could have implications for the re-colonization of abandoned areas through juvenile dispersal and dispersal of beaver from burned habitats to more suitable habitats).

2. Methods

2.1. Site description

Elk Island National Park is located in the Beaver Hills Region, 40 km northeast of Edmonton, Alberta, Canada (Fig. 1). The Beaver Hills are part of the previously glaciated Cooking Lake Moraine—a hummocky landscape dotted with small ponds and lakes. The 194 km^2 park is surrounded by privately and publicly owned land, much of which is developed for agriculture. Undeveloped land still borders the park in some areas, although all land outside the park is separated at the park boundary by a 2.1-m high fence and, in many locations, roads.

Originally established to protect one of the last remaining herds of elk (*Cervus elaphus*) in the region, the park is also home to moose (*Alces alces*), plains bison (*Bison bison bison*), wood bison (*Bison bison athabascae*), white-tailed deer (*Odocoileus virginianus*), and mule deer (*Odocoileus hemionus*). To protect these species from venturing onto the neighboring agricultural lands, the park is completely fenced and, with the exception of coyotes (*Canis latrans*), lacks any resident species of large predators. Good forage and few predators have produced high year-round ungulate densities (13 ungulates/km^2). Parks Canada controls ungulate populations through a regular culling program that incorporates the removal of live elk and bison from the park. Past studies found that unusually high densities of ungulates in EINP have reduced shrub heights and dramatically altered vegetation community composition relative to areas outside the park where ungulate herbivory is significantly lower (Bork et al., 1997a,b; Hood and Bayley, unpublished data).

By the 1880s fur traders had extirpated beaver (*C. canadensis*) from the area that was to become Elk Island National Park (Lin, 1967). A viable population of beaver was re-established in 1941 from beaver that were translocated from Banff National Park. Beaver reached their peak population in 1989 (348 active lodges) and subsequently declined to a 2002 population of 152 active lodges (Fig. 2).

Provincial Highway 16 bisects the park (70% of the park is to the north of Highway 16 and 30% of the park to the south). Most of the park lies within the southern limits of the boreal mixed-wood forest (formerly considered to be part of the Aspen Parkland Natural Subregion; Achuff, 1994). Trembling aspen (*Populus tremuloides*) is the dominant tree species in the park, although balsam poplar (*Populus balsamifera* L.) and white birch (*Betula papyrifera* Marsh) occur in moist areas. Pockets of black spruce (*Picea mariana* Mill.) and white spruce (*Picea glauca* (Moench) Voss) also occur, but are more common in the northern part of the park. Shrub understory in the deciduous forests is diverse and includes beaked hazel (*Cornus cornuta*), serviceberry (*Amelanchier alnifolia*), choke cherry (*Prunus virginiana*), pin cherry (*Prunus pensylvanica* L.f), willow (*Salix* spp.), prickly and wild red rose (*Rosa acicularis* Lindl. and *R. woodsii* Lindl.), raspberry (*Rubus idaeus*), highbush cranberry (*Virburnum edule* Michx.), red osier dogwood (*Cornus stolinifera* Michx.), snowberry (*Symphoricarpos occidentalis* Hook.), currant (*Ribes* spp.) and honeysuckle

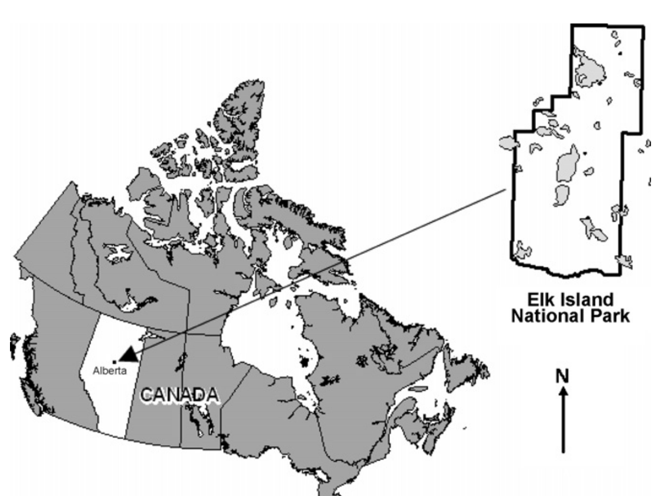

Fig. 1. Location of the 196 km^2 Elk Island National Park, Alberta, Canada. Shaded areas in inset map are larger water bodies within the park.

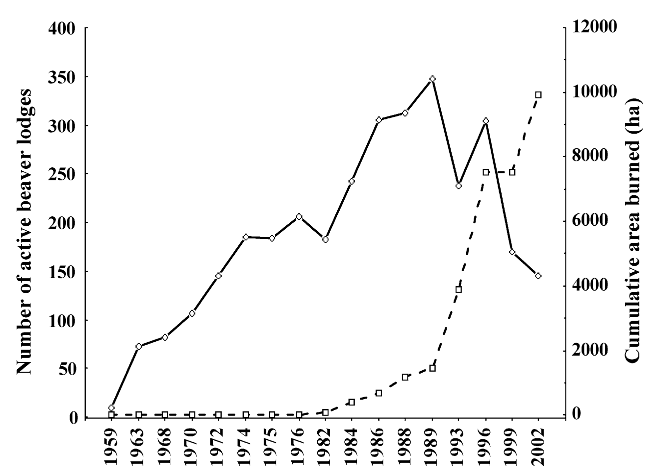

Fig. 2. Number of active beaver lodges from 1959 to 2002 and cumulative area burned through prescribed fire in Elk Island National Park, Canada. Solid line indicates number of lodges occupied by beaver and dotted line indicates the cumulative area (ha) burned since 1979.

(*Lonicera* spp.). Beaver in the park use many of these tree and shrub species as forage and construction material (Skinner, 1984; Hood and Bayley, unpublished data).

2.2. Fire history and beaver lodge occupancy data

All fires since 1979 were mapped and entered into the EINP Geographic Information System (GIS) fire history database. Associated data included date, location, and size of each burn. Parks Canada did not consistently collect data relative to burn severity. There have been 121 prescribed fires in the park since 1979 and all but 6 (in October 1991) were lit between April and June. Fires ranged in size from less than 1 ha to 1,059 ha. Some areas of the park have not been prescribe burned (49% of the park); others have been burned repeatedly (Fig. 3). Although a fire history study has not been done for the park, the typical fire cycle in the Canadian boreal forest region is 75–100 years depending on the local fire regime (de Groot et al., 2003). Parks Canada has burned the equivalent burn area of the size of the entire park over one full fire cycle since the park was established in 1906 (Otway, 2005). The largest burns have occurred in the past 20 years. Currently the Beaver Hills Initiative is pursuing a fire history study for the park and surrounding area (Otway, 2005).

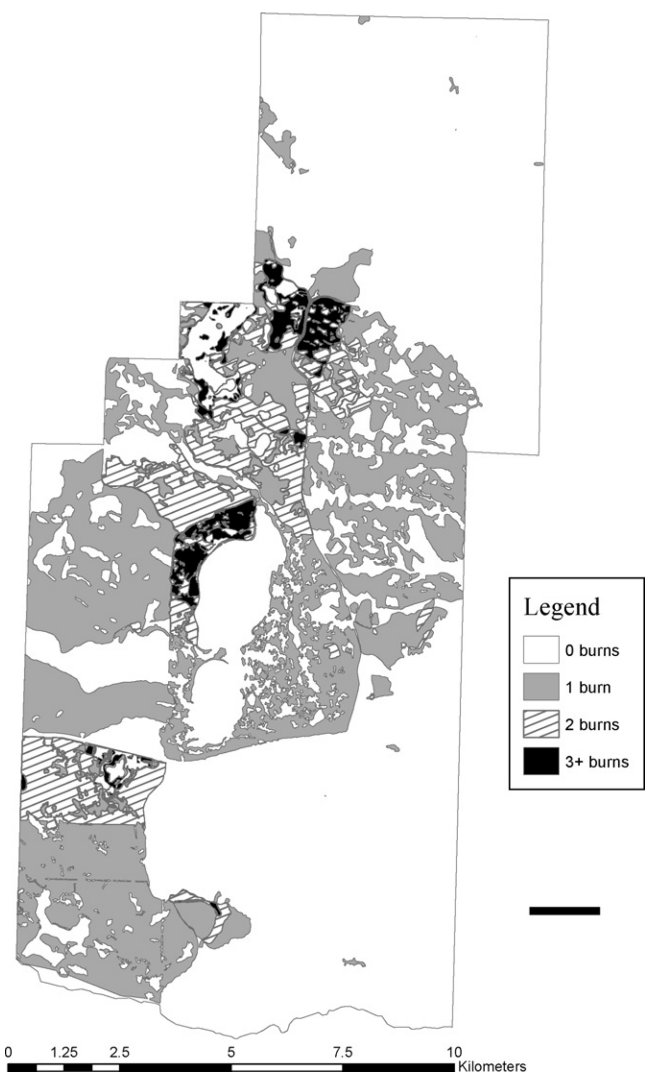

Fig. 3. Fire frequency (1979–2003) in Elk Island National Park, Canada.

Long-term occupancy data for beaver lodges in EINP indicate that beavers (1) construct new lodges, (2) restore abandoned ones, and (3) in some areas occupy the same lodge for more than a decade. Beaver in EINP almost exclusively live in lodges and bank dens are extremely rare. Park-wide helicopter and/or ground surveys of fall beaver lodge occupancy have been conducted by Parks Canada every 1–3 years since their reintroduction. During these surveys, all lodges within the park are mapped and assessed for occupancy status and aerial survey data are confirmed with site visits. A lodge is considered active if a fall food cache is present, a method consistently used in the park since the 1940s. Food caches, winter stores of branches and twigs adjacent to the main lodge entrance, are apparent from the both air and the ground. For lodges examined in this study, lodge occupancy data were consistently available for the year prior to and immediately following a burn as well as for every 1–3 years thereafter. Since 1989, all lodges (active and inactive) have been entered into a GIS where they were each given a unique identifier so their occupancy status can be tracked over time.

Although Parks Canada has conducted regular beaver lodge surveys in EINP, observers did not consistently label individual lodges until 1989. As a result, all beaver lodges that were either visible in, or were constructed after 1989 have excellent data relative to their long-term trends in occupancy. For this reason we used the post-1989 data to assess occupancy trends relative to fire. By 2001, the end of our survey period, this dataset included 734 lodges.

We used two GIS databases (fire history and beaver lodge occupancy) to develop a complete fire history for each lodge (e.g., burned versus unburned, burn frequency, year burned, time since last burned). A lodge was considered to be influenced by fire if the lodge was completely within a burn polygon in the GIS fire history database. If the lodge was outside the polygon we classified it as ‘‘unburned’’ because the beaver colony would have access to adjacent unburned habitats. Using occupancy data, we classified lodges in the GIS as either active (‘‘1’’) or inactive (‘‘0’’) for each year since 1989.

2.3. Wetland coverage within burned and unburned areas

To ensure that differences in beaver lodge occupancy in burned and unburned areas were not confounded by the availability of wetland habitats; we determined the areas of all the 1920 wetlands in the burned ($n = 814$) and unburned areas ($n = 1106$). Data were first tested for homoscedascity using the Brown and Forsythe test (Brown and Forsythe, 1974) and normality using the Shapiro–Wilk's W test (Shapiro et al., 1968). We then used a t-test for independent samples to compare differences in the areas of wetland between burned and unburned areas ($\alpha = 0.05$).

2.4. Preliminary data analysis for effect of fire on lodge occupancy

To obtain an initial assessment of the effect of fire on lodge occupancy from 1989 to present, we pooled occupancy data for areas burned only once, and compared the percent occupancy immediately prior to and after a fire. We tested for significance using the Cochran Q test (Bishop et al., 1975). Cochran Q test is an extension of the McNemar's χ^2-test and is useful for dichotomous data (Zar, 1999; Cochran, 1950). In the case of our study, we tested whether lodge occupancy was associated with whether or not the area had been burned.

To assess the effects of burning on lodge occupancy over time, we also tracked individual years following a burn to examine lodge occupancy rates for up to 12 years following the burn. There were 4 years when fires were lit for which there was a large enough sample size to track lodge occupancy over time (1991, 1994, 1995, and 2000). To avoid the confounding factor of multiple burning, only lodges that were in areas burned once were considered in this initial analysis. We then used least squares regression for each of the 4 fire years to determine whether precipitation was associated with the magnitude of decline in the number of active lodges for the years following a burn. We also used Pearson r correlation to test the correlation between the pattern of lodge occupancy in burned and unburned areas over the same time period (StatSoft Inc., 2003).

2.5. Model development and analysis

Further analysis examined the influence of several variables that potentially affected lodge occupancy (including the number of burns). We used the 1989–2001 beaver survey data to analyze effects of fire on beaver lodge occupancy over a 12 year period because this data set included all lodges surveyed since 1989 that were active at least once during the survey period ($n = 734$). All lodges that had been permanently abandoned prior to 1989 were removed from the analysis.

Because of the binomial nature of the response variable (lodge occupancy), we used multiple logistic regression to model beaver lodge occupancy relative to burn frequency, years since the lodge area was last burned, and distance of each lodge to the nearest active lodge (StatSoft Inc., 2003). The first two variables accounted for aspects relative to fire history, while the third acted as a spatial variable that provided a context for possible dispersal distances and how fire might affect the overall density of occupied beaver lodges. We used Akaike's Information Criterion (AIC) for model selection (Burnham and Anderson, 1998) and evaluated candidate models using model weights (ω) and ΔAIC. Models with $\Delta\text{AIC} > 2$ of the best model were considered worthy of future consideration (Burnham and Anderson, 1998). All statistical tests were evaluated using Statistica (StatSoft Inc., 2003) and R statistical software (R Development Core Team, 2005).

Initially we analyzed full models that included the predictor variables fire frequency + years since last burn + distance to the nearest active lodge and their interaction effects (Table 1).

Table 1
Description and representation of a priori models used for relating effects of number of burns (frequency), time since last burn (years), and distance to nearest active lodge (distance) on occupancy of beaver lodges (status) in Elk Island National Park

Model	Description
Frequency	Frequency alone
Frequency + years	Additive frequency and years effects
Frequency + distance	Additive frequency and distance effects
Frequency + years + distance	Additive frequency, years and distance effects
Frequency + years + distance + frequency × years	Additive frequency + years + distance effects with a frequency × years interaction
Frequency + years + distance + frequency × distant	Additive frequency + years + distance effects with a frequency × distance interaction
Frequency + years + distance + distance × years	Years alone
Years	Years alone
Distance	Distance alone
Status	Constant status

Data were developed from digital GIS data (e.g., fire history data for EINP) and beaver lodge occupancy data.

These models were chosen to test fire-specific variables (fire frequency and years since last burn) as well as a variable that accounted for lodge occupancy independent of fire (distance). During the initial model runs any variable that was highly correlated ($R > 0.7$) and not significant on its own was removed from the final analyses.

3. Results

3.1. Water coverage within beaver management units

Over 33% of the park is covered by wetlands (Hardy and Associates, 1986) represented by lakes (15%), marsh (11%), swamps (3%), ponds (2%), fens (1%), and bogs (1%). The park lacks any major rivers and streams are generally shallow and intermittent. Burned and unburned areas had similar wetland coverage ($P = 0.11$). Average wetland area was 2.49 ha $\pm$ 14.28 S.D. in burned areas (range = 0.05–393.17 ha) and 4.09 ha $\pm$ 26.01 S.D. (range = 0.04–559.78 ha) in unburned areas. The largest lake in the park, Astotin Lake (559.78 ha), was burned along its west and south shores, but because it was not completely surrounded by a burn, its water area was classified as being in an unburned area. Despite inclusion of this large water body into the unburned class, average wetland size in the burned and unburned areas was not different ($P = 0.11$).

3.2. Preliminary effect of fire on beaver lodge occupancy

For lodges burned only once, lodge occupancy for the year immediately prior to a burn was significantly higher (41.1%) than the year immediately after a burn (25.38%, $Q = 10.4716$, d.f. = 1, $P = 0.001$). Although populations in some of the burned areas did increase in the years after burning, they never recovered to their original occupancy levels over the 12-year period we examined (Fig. 4). Total annual precipitation did not significantly affect the number of active

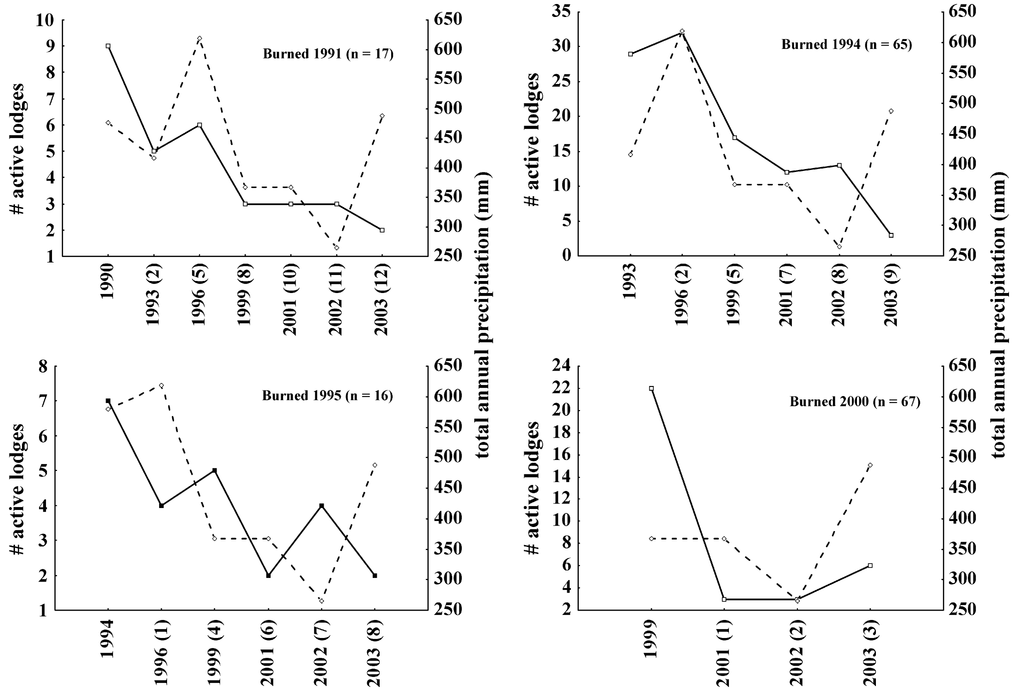

Fig. 4. Beaver lodge occupancy during 3–12 years following burns conducted in 1991, 1994, 1995, and 2000 (number of years post-burn are in brackets following the year). The solid line (—) indicates the number of active lodges, and the dashed line (- - -) indicates the total annual precipitation (mm) during the same period. Only lodges burned once are included in the graph.

beaver lodges during any of the 4 fire years that we examined ($R^2 < 0.22$ and $P > 0.29$ in all cases). Although occupancy in the unburned areas in 1991 ($R = 0.98$, $P < 0.001$) and 1994 ($R = 0.89$, $P < 0.02$) showed similar patterns of lodge occupancy to those in burned areas, this trend was not observed in 1995 ($R = 0.34$, $P = 0.51$) and 2000 ($R = 0.04$, $P = 0.96$; Fig. 5).

3.3. Model results for the effects of fire on beaver lodge occupancy

The lowest AIC was for the model that included frequency only (Table 2). However, differences in AIC values between the model that included frequency and the model that included both frequency and other variables were less than $\Delta AIC > 2$. We

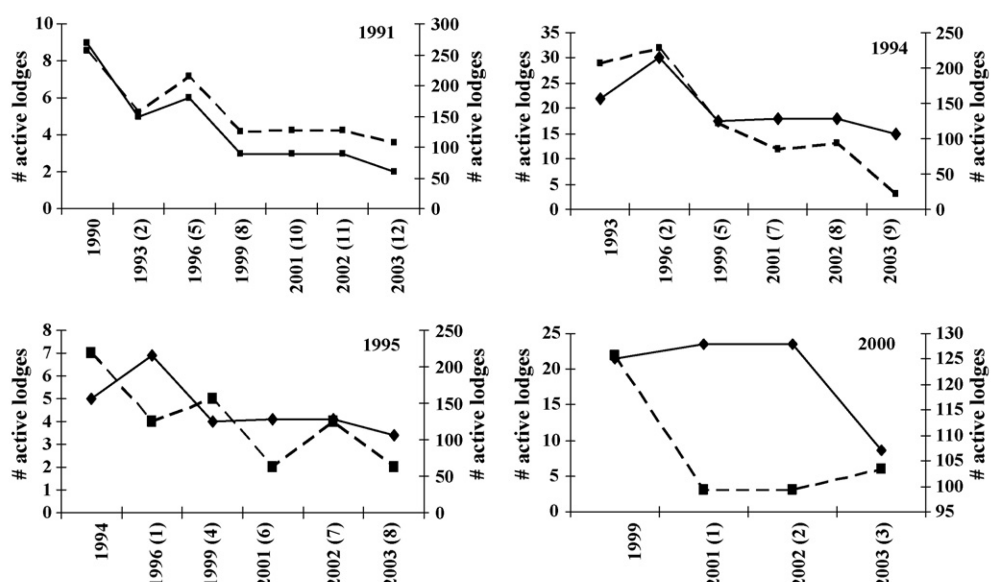

Fig. 5. Beaver lodge occupancy during 3–12 years following burns conducted in 1991, 1994, 1995, and 2000 (number of years post-burn are in brackets following the year). The solid line (—) indicates the number of active lodges in unburned areas, and the dashed line (- - -) indicates number of active lodges in burned areas during the same period. Only lodges burned once are included in the graph.

Table 2
A comparison of models used for predicting occupancy of beaver lodges relative to number of burns ("frequency") and the distance to the nearest active lodge ("distance") in Elk Island National Park (EINP), Alberta

Model	−2 LL	K_i	AIC	Δ_i	w_i
Frequency	−326.41	2	656.82	0	0.680
Frequency + distance	−326.17	3	658.33	1.51	0.319
Distance	−333.07	2	670.13	13.31	0.001

Values include −2 log likelihood scores (−2 LL) and number of parameters (K). The variable "time since the last burn" was highly correlated with "frequency" ($R = -0.78$, $P < 0.001$) and therefore removed from the final analysis. We assessed models by ranking AIC values (Δ_i) and weights (w_i), which describe the likelihood of the model.

removed years since last burn from the analysis because this variable was highly correlated with frequency ($R = -0.78$, $P < 0.001$), and was not a good predictor of lodge occupancy on its own. When years since last burned was included in the model, the β-coefficient for the frequency of burns was −0.86 ($P = 0.005$)and 0.06 ($P = 0.197$) for years since last burned (Table 3). When years since last burned was removed from the model, the β-coefficient for the frequency of burns was reduced to −0.58 ($P = 0.001$). The β-coefficient for distance to nearest active lodge did not show any significant change ($\beta = -0.002$, $P = 0.49$ for both models). When the variable "years since last burned" was removed, burn frequency alone accounted for 0.68 of the AIC weight (Table 2).

According to the model with only burn frequency as a variable, the odds of a lodge being active decreases by 58% for each additional burn. This is evident in the data: after three burns lodges were often permanently abandoned (Fig. 6). The β-coefficient was the same for the model with frequency alone and the model that included frequency and distance. Although not significant, the odds of a lodge being active decreases by 0.1889 for each additional kilometer a lodge is from its nearest active lodge (Table 3).

4. Discussion

4.1. Fire frequency and timing impacts

As predicted, occupancy of lodges by beavers was significantly lower in areas that had been burned than in unburned areas. There was a marked decrease in lodge occupancy in areas that have been burned more than once. Despite the fact that regeneration of woody plant species and, therefore an increase of forage for beaver, are often predicted after a burn (Bartos and Mueggler, 1981; Bailey and Whitham, 2002), there did not appear to be a trend in lodge reoccupation in years following a fire.

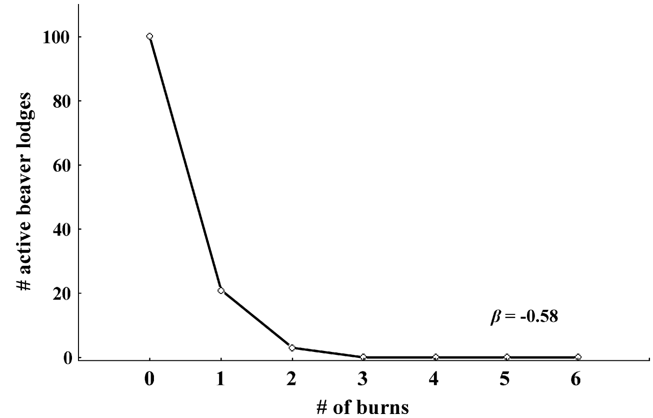

Fig. 6. Beaver lodge occupancy during 2001 in Elk Island National Park relative to the number of burns the lodge was exposed to since the start of the prescribed fire program in1979.

Fire frequency has long been understood to regulate grassland ecosystems. In some agricultural systems, a combination of repeated burning, grazing, and season of burn is used as a means to eliminate woody vegetation and produce grassland pastures (Anderson and Bailey, 1979; Bailey et al., 1990). Historians attribute the combination of aboriginal and natural fires and herbivory to the resilience of the prairie grasslands of pre-colonial North America (Lewis, 1982; Campbell et al., 1994). Fire frequency shorter than the time required for plants to reproduce can limit regeneration of plants whose seeds are do not require fire to aid in seed release (de Groot et al., 2003). In Elk Island National Park, the combination of fire and intensive browsing by ungulates appear to have increased grassland habitats to the detriment of beaver. The park has been managed under the assumption that the fire regime is similar to other boreal systems (75–100 years; de Groot et al., 2003). In those systems one assumes beaver and fire have successfully coexisted, although no empirical data exist. As the park obtains more site specific fire history data and can adapt its management to more natural fire cycles, it will be interesting to see if beaver populations are less impacted than under the current fire management practices.

Time of year for the burn may also play a role at EINP. Season of burn is related to tree mortality and the ability for a

Table 3
Estimated coefficients and 95% confidence intervals (CI) for logistic regression models assessing the effects of the number of burns ("frequency"), the distance to nearest active lodge ("distance"), and the years since last burn ("years") on beaver lodge occupancy

Variable	Frequency + distance + years				Frequency + distance				Frequency			
	β	exp(β)	95% CI for exp(β)		β	exp(β)	95% CI for exp(β)		β	exp(β)	95% CI for exp(β)	
			Lower	Upper			Lower	Upper			Lower	Upper
Frequency	−0.86	0.42	0.23	0.77	−0.58	0.56	0.4	0.8	−0.58	0.56	0.4	0.8
Distance	−0.16	0.85	0.5	1.46	−0.19	0.83	0.48	1.42	–	–	–	–
Years	0.06	1.01	0.97	1.17	–	–	–	–	–	–	–	–
Constant	−1.39	2.66			−1.29	0.27			−1.39	0.25		

tree to resprout (de Groot et al., 2003). Trees are more likely to be girdled by fire prior to green-up of ground cover and after leaves cover the ground in the fall. In North America, most natural fires occur in the summer, a scenario not replicated by prescribed burning programs. Most prescribed burns in EINP during our study period were between April and June. Higher moisture levels in the spring typically allow for safer burning conditions. Fall burning can present greater public safety hazards due to lower moisture levels. Burning in the fall can also dramatically reduce abundance of grasses and woody plants (Bailey et al., 1990), the winter food supply for many herbivores. Unfortunately, early spring is also the time when most emergent wetland vegetation (e.g., *Typha latifolia*) is dry and flammable. In late spring, summer or early fall these obligate wetland species are green. Early spring burning in wetland habitats results in fires that extend to the water's edge and, in the case of beaver ponds, right up to and, on occasion, over any adjacent lodges. The impact to beaver lodges may be another cause of the reduced occupancy after a burn and after repeated burns. Buffering water bodies from prescribed fire within EINP could help maintain habitat quality for beaver and reduce direct losses of beaver lodges and dams.

The combination of drought and fire had a devastating effect on beaver in EINP in 2000. After a particularly large hot fire that was lit in 2000 (approximately 800 ha), the number of active beaver lodges in the burned area declined from 15 active lodges to 3 active lodges over the course of a year. Lodges in this area that were active before the fire have not yet been reoccupied (as of 2006). It is possible that habitats already compromised by drought were not capable of mitigating the effects of burning.

4.2. Ungulate herbivory

Unlike many areas where prescribed fire is used for vegetation management, effects of prescribed fire in EINP are confounded by an unusually high year-round ungulate population. Bork et al. (1997b) found that mature forms of *A. alnifolia* and *P. viriginiana* were found only outside the boundaries of EINP, where fire and high levels of ungulate herbivory were non-existent. This finding is despite the fact that *A. alnifolia* is a fire tolerant species (Noste and Bushey, 1987). Although the presence of *C. cornuta* and *R. idaeus* were positively correlated with burn intensity inside the park, they were significantly shorter and more hedged than those same species found outside the park (Bork et al., 1997a; Hood and Bayley, unpublished data).

Overgrazing by ungulates in burned areas might inhibit plant growth to the point of preventing regeneration of adequate beaver forage. Studies indicate that ungulates are often drawn to areas shortly after a burn (Vinton et al., 1993; Bailey and Whitham, 2002) and the associated increase in foraging by ungulates in burned areas would delay the regeneration of woody plants. An alternative hypothesis is that beaver have a natural cycle in which they abandon areas where food resources have been exhausted after years of beaver foraging (Ives, 1942); however this hypothesis would also assume a similar pattern of lodge abandonment in both burned and unburned habitats, which was not observed for all years.

4.3. Dispersal and foraging distances

Distance did not have a significant influence on occupancy. Along with vegetation patterns, topography often dictates how beaver move across the landscape (Johnston and Naiman, 1990a). Regardless of how much area is burned, beaver will always be limited by other environmental factors such as climate, predation, the proximity of suitable water bodies, and beaver density. Such limits are reflected in the varying distances to active lodges found from year-to-year in both burned and unburned areas and might explain the lower β-coefficient and higher AIC value of distance in our models.

If an area is burned extensively, however, beaver must increase their foraging distances to find unburned vegetation if they wish to remain in the area. As central place foragers, any increase in foraging distance for beaver would increase their exposure to predation as well as the energy required to bring food and construction materials back to their lodge (Fryxell and Doucet, 1991). Conversely, if they chose to move to another wetland, they risk predation and extensive search times, particularly if suitable wetlands are not close by. This aspect of dispersal and foraging in burned areas warrants further study.

Fire is a natural process in many ecosystems and has been used as a management tool in North America since pre-Columbian times (Lewis, 1982). Its current use extends to many continents and various ecosystems. Repeated use of fire, especially in grazing systems, controls and suppresses woody vegetation (Bork et al., 1997b; Roques et al., 2001). Although maintenance of grasslands is beneficial for many grazing herbivores, herbivores that rely heavily on woody plants could be excluded from these systems. Unlike beaver, ungulates are capable of moving across landscapes in search of better forage and therefore are more resilient to localized disturbance such as fire. When beaver habitat is depleted or destroyed, beaver must disperse to a new central location before they resume their foraging and food caching activities.

The long-term reduction in occupancy after repeated burning supports our prediction that multiple fires reduce the number of active beaver lodges. In addition, the interactive effect of fire and drought brings to question whether extensive burning reduces the landscape's ability to mitigate the effects of severe drought in wetland habitats, especially when coupled with high levels of herbivory. Further research is required to fully understand the cumulative impacts of multiple perturbations on beaver habitat and populations including examination of factors influencing re-occupancy after fire.

5. Management implications

Use of prescribed fire is often a necessary and irreplaceable tool in the management of natural areas. After a century of fire suppression in many areas of North America, the resultant fuel loading has created an unacceptable risk to public safety. Loss of landscape heterogeneity has reduced habitat diversity, with subsequent impacts on ecological processes (Woodley, 1993; White et al., 1998). However, prescribed fire must be conducted in ways that sustain existing ecological processes and key species that have long-term benefits to larger ecosystems. In particular, repeated burns in beaver habitat should be avoided and burning wetland areas during times of drought is not recommended.

Beaver are considered a keystone species (Paine, 1966; Johnston and Naiman, 1990b) and their wetland building activities provide habitats for many other organisms. In the context of extremely high density of ungulates, repeated burns in beaver habitat appear to have caused a dramatic decrease in lodge occupancy (and therefore population). Fire could be important for the regeneration of many woody plants species used by beaver, but these benefits would depend on several factors such as herbivory, fire frequency, and fire severity. Faced with predictions of an increasingly dry climate on the Canadian prairies (Hogg and Bernier, 2005), the potential negative impact of fire becomes even more significant. The challenge then, is to broaden our understanding of how beaver and fire coexist in concert with other species competing for the same resources in changing climatic conditions, and apply that knowledge to current management practices to maintain a broad range of species in ever changing environments.

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